EXPERIMENTAL VERIFICATON OF DE-MORGAN'S THEOREM

LAB 3

CSE-202L DIGITAL LOGIC GATES

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Section C

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OBJECTIVES:

- To experimentally verify De-Morgan's theorem.
- To simplify complex circuits by applying De-Morgan's theorem.

COMPONENTS:

- Battery
- Jumper wires
- 3, 1K Ω resistors
- Dip switch
- IC's; 7400, 7402, 7404, 7408, 7432

TASK 1

VERIFYING DE-MORGAN'S THEOREM:

1ST THEOREM:

$$\overline{(X+Y)} = \bar{X}.\bar{Y}$$

Left side of the equation is expression for NOR gate.

Right side of the equation is expression for bubbled AND.

PROCEDURE:

- TESTING SWITCHES AND GATES:
  - We will start by checking switches and gates of the ICs to see if they work properly.
  - This is done by inserting the DIP switch on the breadboard, and connecting two resistors with the switch being tested.
  - One resistor is connected to 9V power rail, and the other is grounded.
  - The output is taken from the switch and passed to positive lead of an LED. The negative lead is grounded.
  - If the LED turns on at 1, and turns off at 0, the switch is fine.

Connections are shown in fig 1, in the section below, "Demonstrating lab Performance".

- MANIFESTING L.H.S:
  - Insert ICs' 7432, 7404, and 7408 on the mid partition of the breadboard.
  - Install the DIP switch. Power the switch at 9V by inserting one end of the 1K resistor at 9V rail, and the other end at the corresponding column (in which the pin of the switch is inserted). Similarly, take another resistor and insert its one end at the column of the switch, and ground the other end.

- Take input for the IC 7402 from the switch. To do this, insert two jumper wires at the output of any two switches and connect the other ends at the columns of the input for the NOR gate.
- Now, take the output of the gate by inserting a jumper wire at the column for the output, and pass it to the positive lead of the LED. Ground the negative lead.

Connections are shown in fig 2.

- MANIFESTING R.H.S:
  - Take inputs X, and Y from the two switches.
  - Pass the input to any two gates of the 7404 IC through jumper wires.
  - Take the outputs from the two gates of 7404 IC, and pass them to any gate of the 7408 IC.
  - Finally, take the output from the 7408 IC and pass it to LED.

Connections are shown in fig 3.

ALGEBRAIC PROOF USING TRUTH TABLE METHOD FOR 1ST THEOREM:

A	B	$\overline{(A+B)} = C$
0	0	1
0	1	0
1	0	0
1	1	0

$\bar{A}$	$\bar{B}$	$\bar{A}.\bar{B}$
1	1	1
1	0	0
0	1	0
0	0	0

2ND THEOREM:

$$\overline{(X.Y)} = \bar{X} + \bar{Y}$$

Left side of the equation is the Boolean expression for NAND gate.

Right side of the equation is expression for the Bubbled OR gate.

PROCEDURE:

- Manifesting LHS:
  - Install DIP switch and resistors as explained above.
  - Install 7400 IC on the breadboard.
  - Give two inputs from the DIP switch to any gate of the IC.
  - Give output to the LED via jumper wire.

Connections are shown in fig 4.

- Manifesting R.H.S:
  - Install 7404, and 7432 on the breadboard.
  - Give inputs to any two gates of the 7404 IC.
  - Take the two outputs from 7404 and pass to any gate of the 7432 IC.

- Pass output of the 7432 IC to an LED.

Connections are shown in fig 5.

ALGEBRAIC PROOF USING TRUTH TABLE METHOD FOR 2ND THEOREM:

A	B	$\overline{(A.B)} = \overline{C}$
0	0	1
0	1	1
1	0	1
1	1	0

$\overline{A}$	$\overline{B}$	$\overline{A} + \overline{B}$
1	1	1
1	0	1
0	1	1
0	0	0

DEMONSTRATING LAB PERFORMANCE

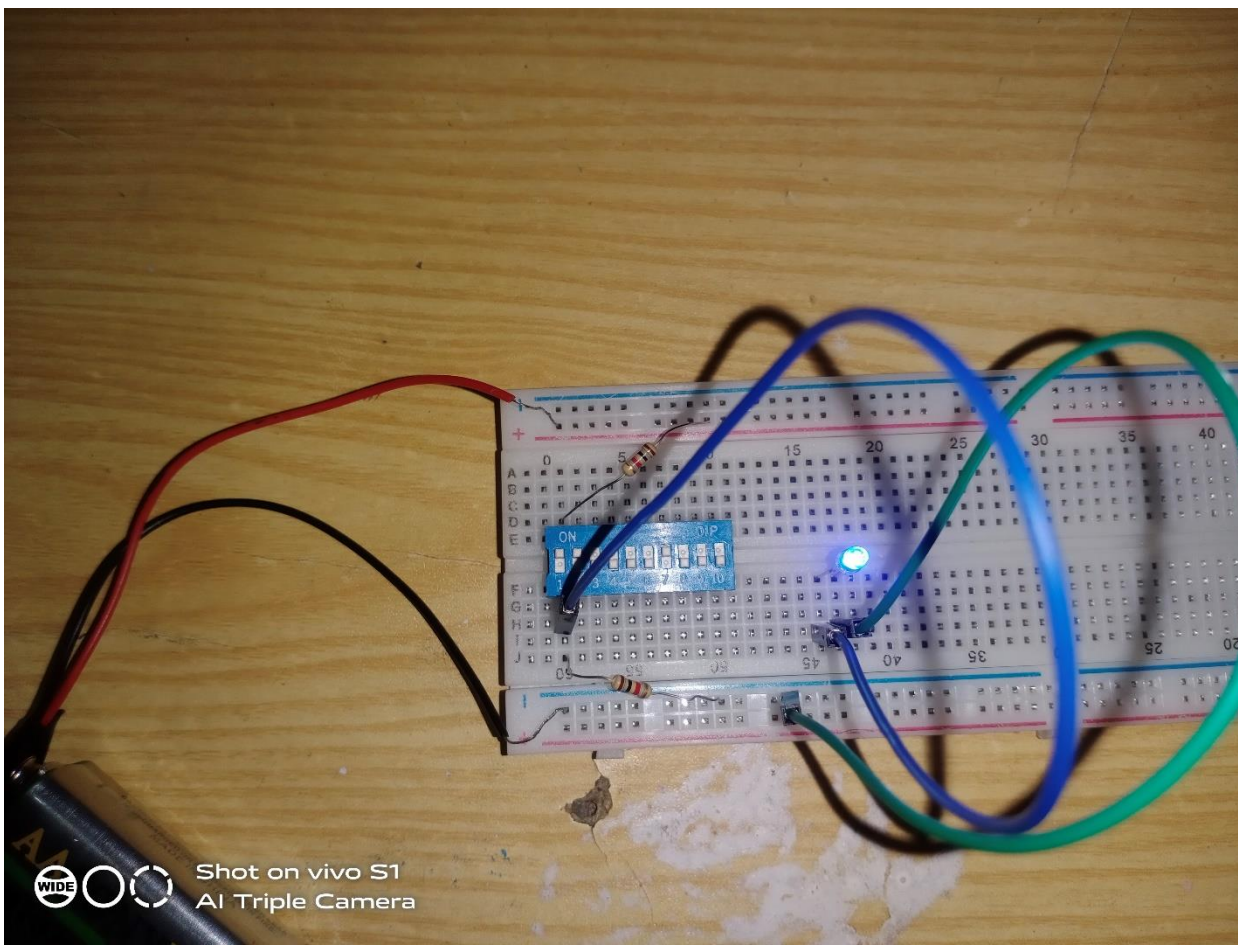


Fig 1

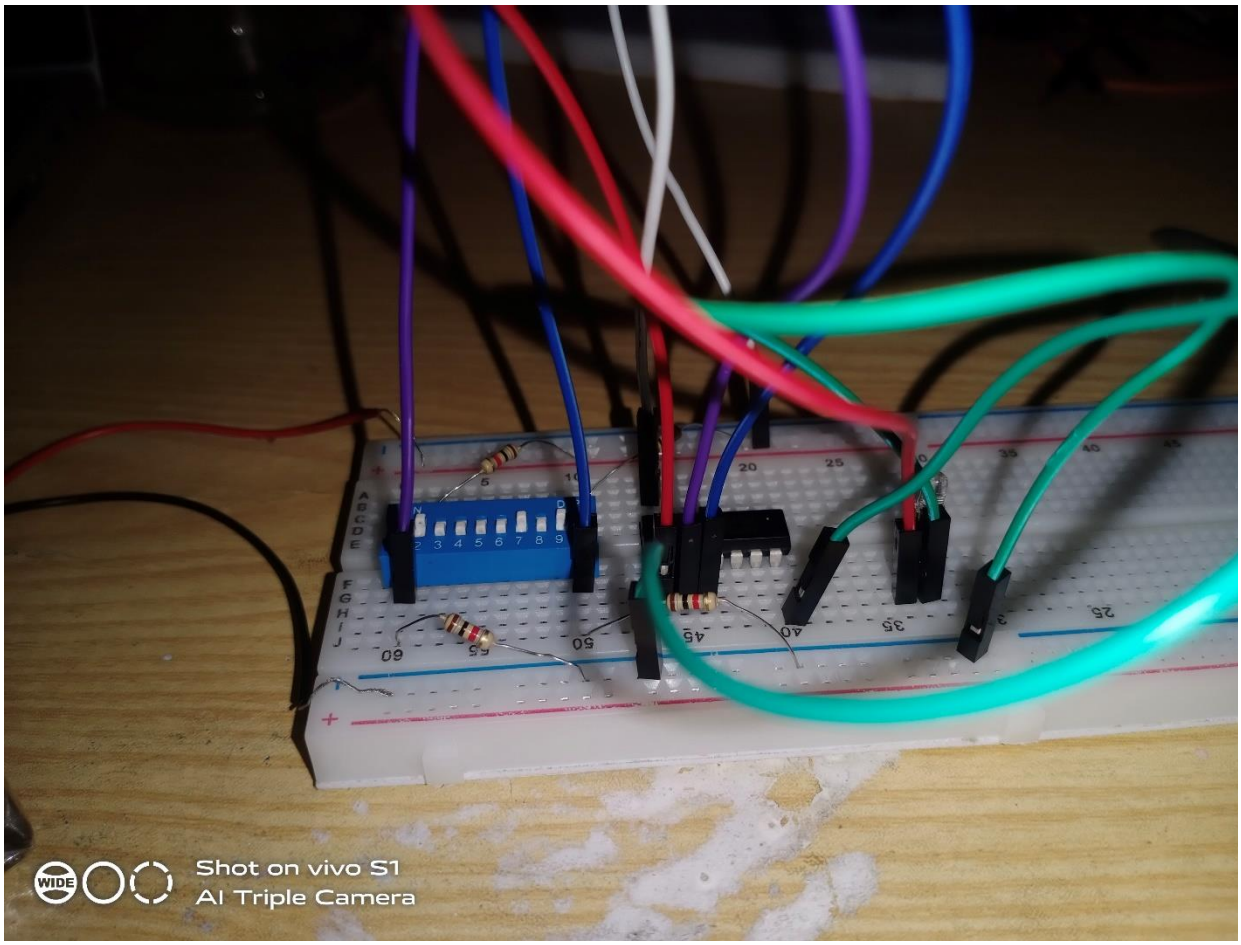


Fig 2

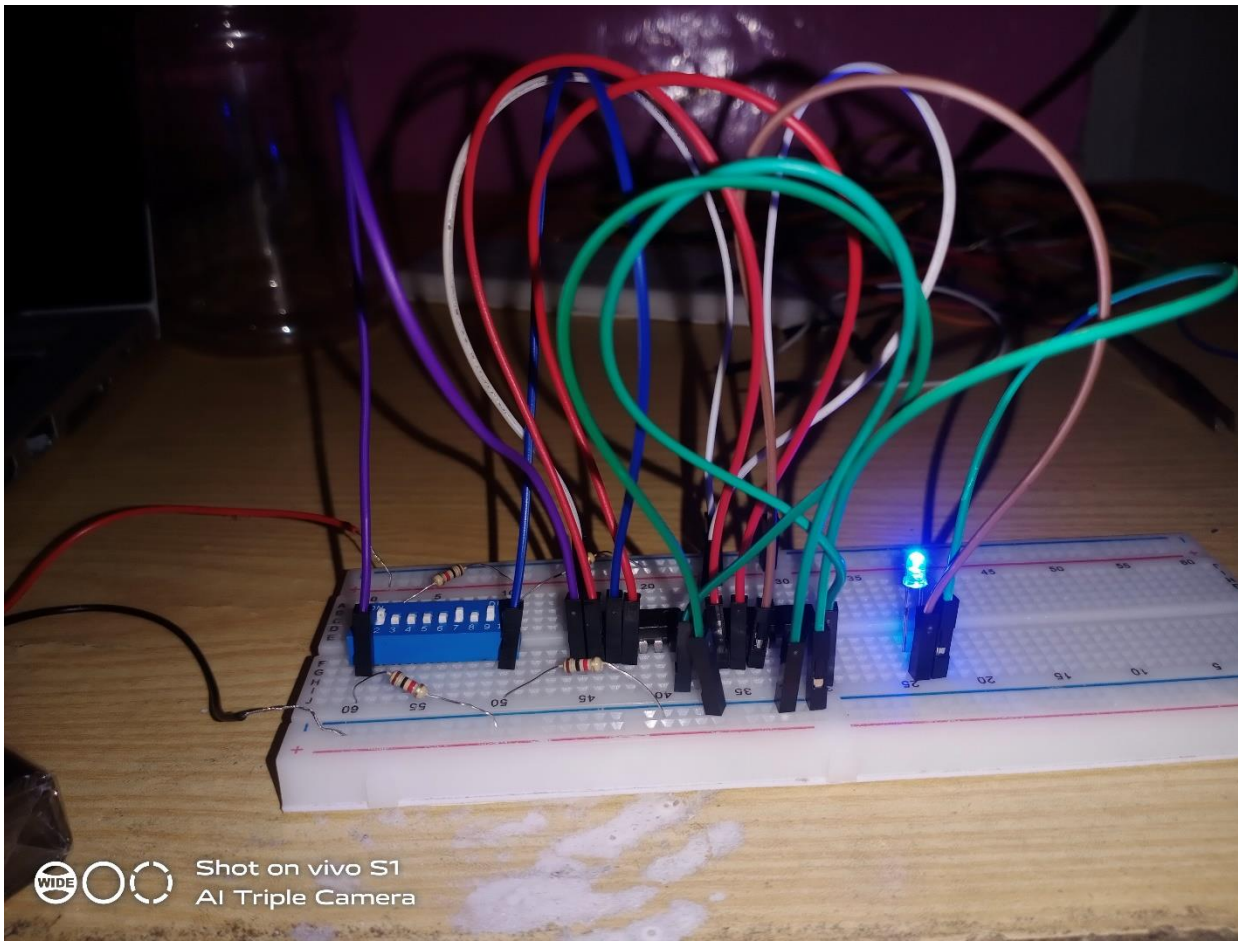


Fig 3

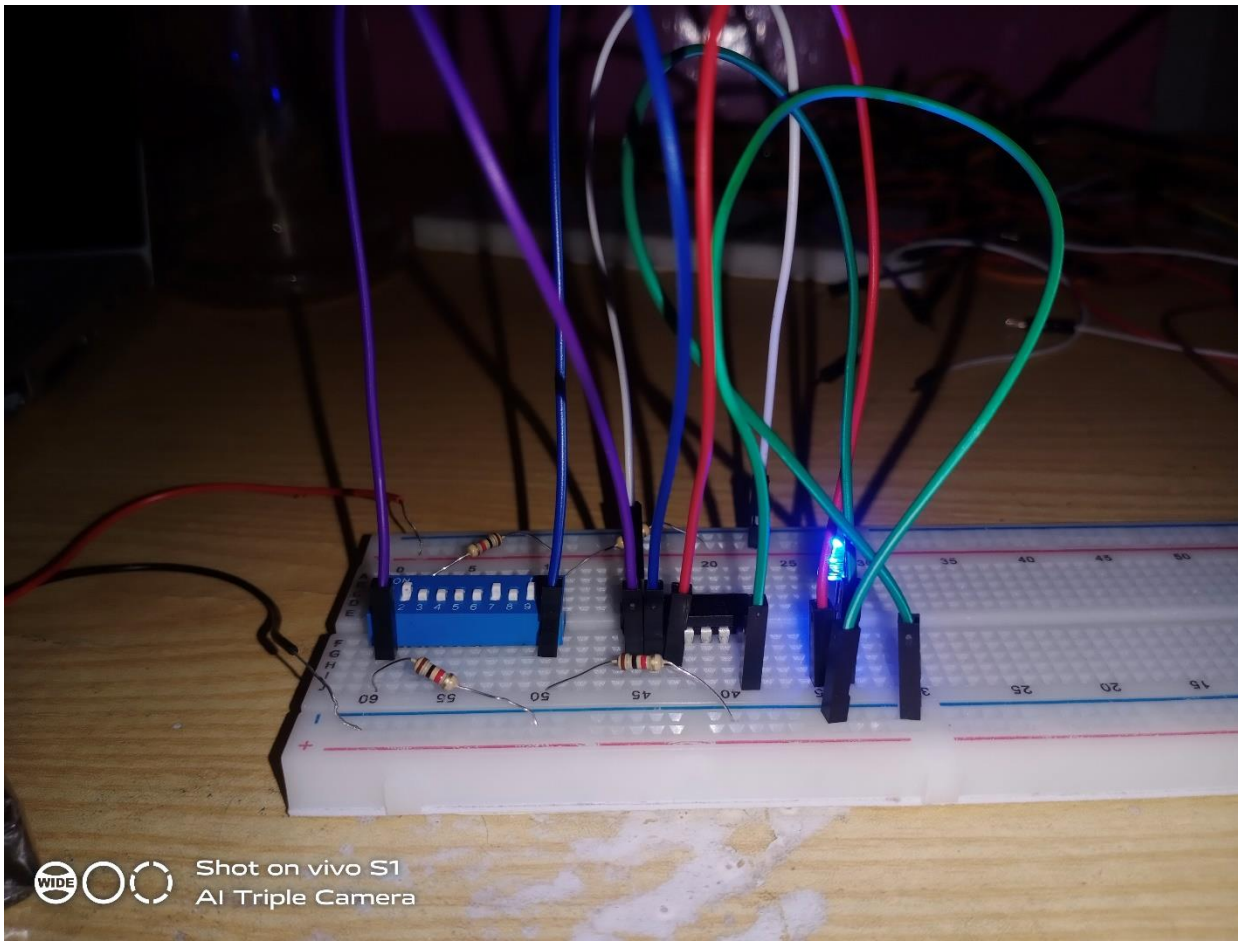


Fig 4

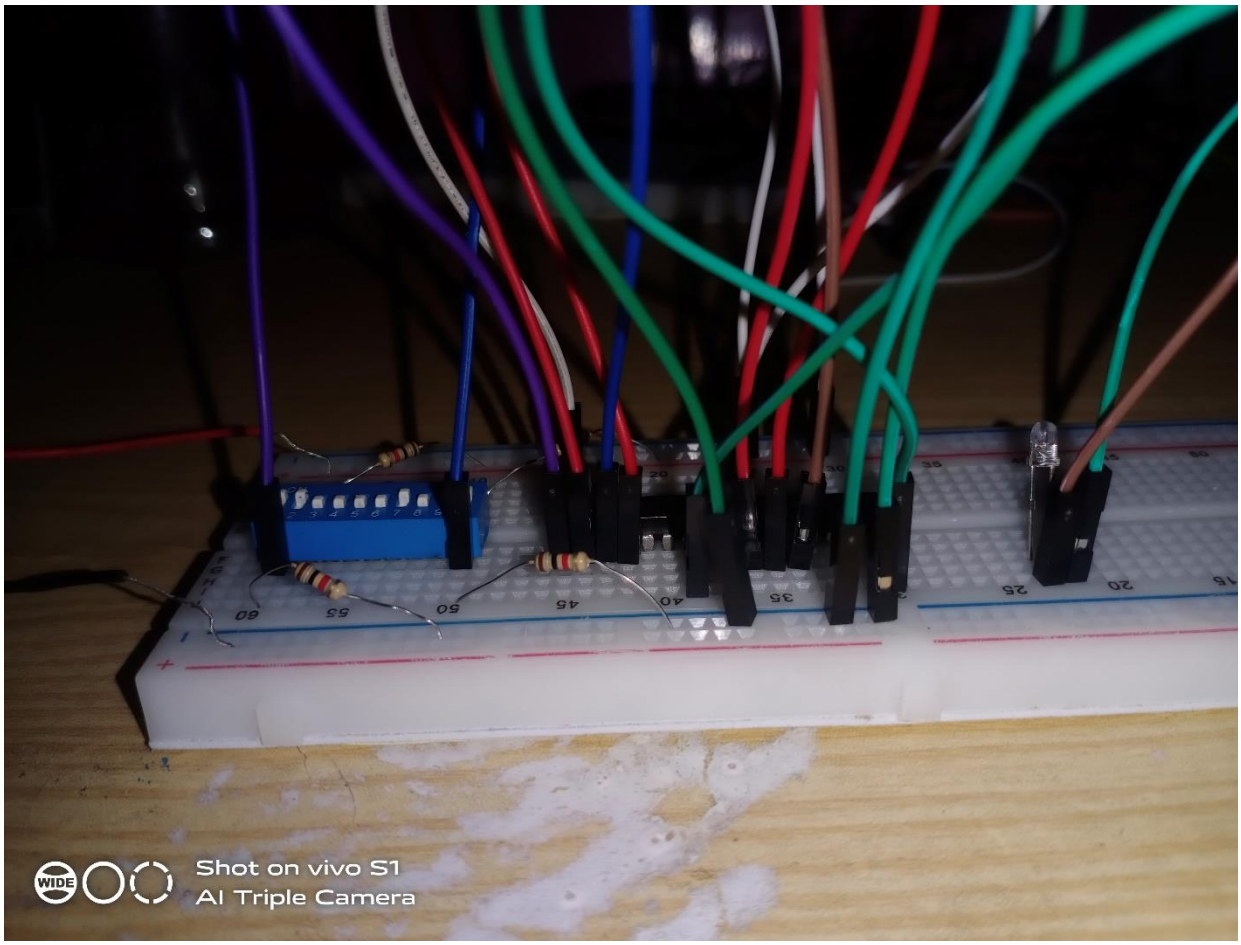


Fig 5

EXPERIMENTAL PROOF USING TRUTH TABLE METHOD FOR 1ST THEOREM:

A	B	$\overline{(A+B)} = C$	$\overline{A}$	$\overline{B}$	$\overline{A}.\overline{B}$
L	L	H	H	H	H
L	H	L	H	L	L
H	L	L	L	H	L
H	H	L	L	L	L

EXPERIMENTAL PROOF USING TRUTH TABLE METHOD FOR 2ND THEOREM:

A	B	$\overline{(A.B)} = C$	$\overline{A}$	$\overline{B}$	$\overline{A}+\overline{B}$
L	L	H	H	H	H
L	H	H	H	L	H
H	L	H	L	H	H
H	H	L	L	L	L

TASK 2: REVIEW QUESTIONS

(1)

- Simplify the expression using De-Morgan's theorems and verify the two expressions experimentally.

F= ((A . B)' + A)'

Applying De-Morgan's Theorem:

Using de-morgan's 1st theorem,

$$F = \overline{(\overline{(A.B)} + A)}$$

$$F = \overline{\overline{(A.B)}} \,.\, \bar{A}$$

- The double bar cancel's out and we are left with,

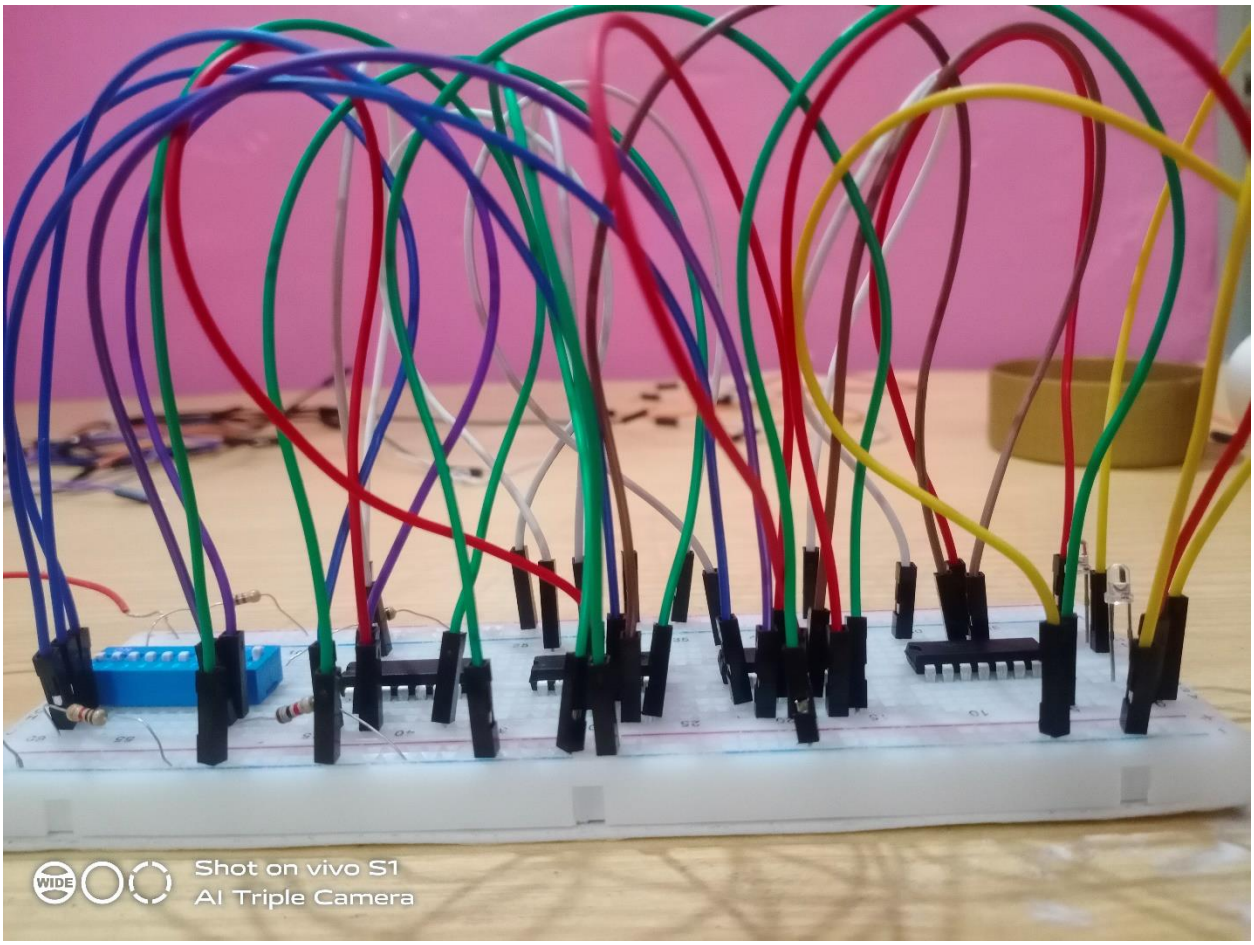
$F = (A.B) \,.\, \bar{A}$, is the simplified form of given expression.

ALGEBRAIC PROOF USING TRUTH TABLE METHOD:

A	B	$\bar{A}$	A.B	$\overline{A.B}$	$\overline{(A.B)} + A$	$\overline{(\overline{A.B}) + A)}$	(A.B). $\bar{A}$
0	0	1	0	1	1	0	0
0	1	1	0	1	1	0	0
1	0	0	0	1	1	0	0
1	1	0	1	0	1	0	0

EXPERIMENTAL PROOF:

A	B	$\bar{A}$	A.B	$\overline{A.B}$	$\overline{(A.B)} + A$	$\overline{(\overline{A.B}) + A)}$	(A.B). $\bar{A}$
L	L	H	L	H	H	L	L
L	H	H	L	H	H	L	L
H	L	L	L	H	H	L	L
H	H	L	H	L	H	L	L



(2)

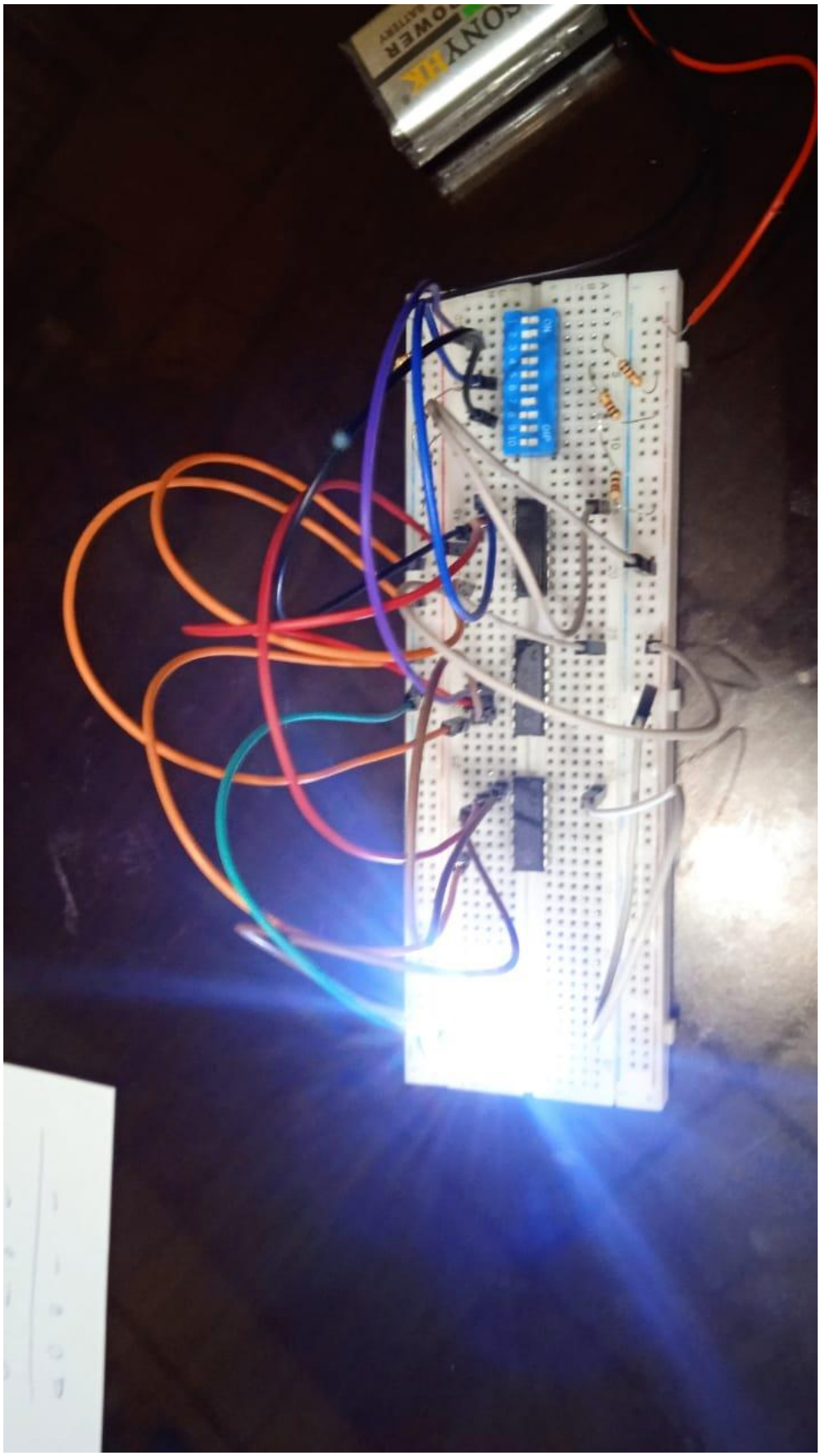
- Determine experimentally whether the given circuits are equivalent. Then use De-Morgan's theorem to prove your answer algebraically.

<u>Expression for 1st circuit:</u> $(A + \bar{B}).\bar{C}$

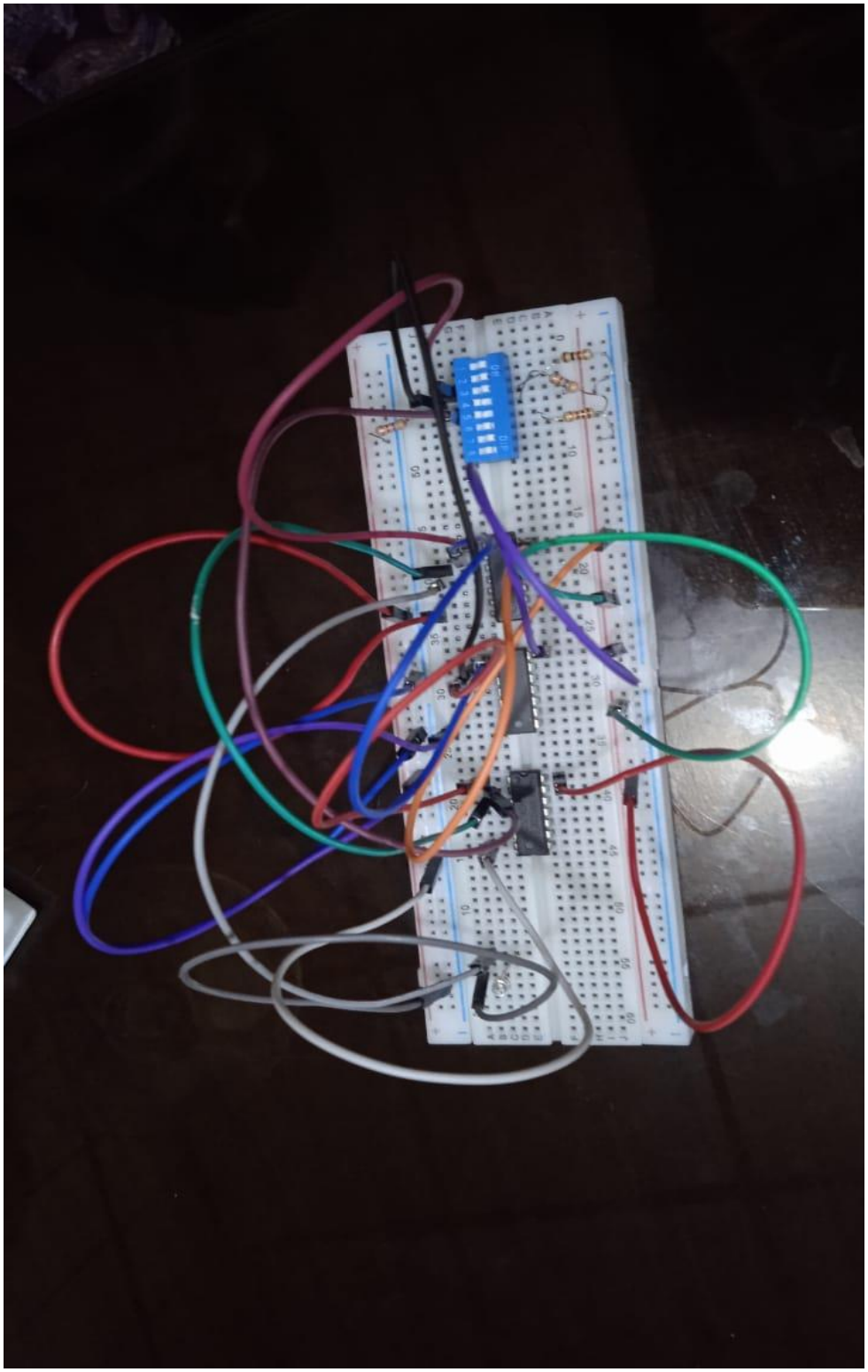
<u>Expression for 2nd circuit</u>: $\overline{((\bar{A}.B) + C)}$

<u>EXPERIMENTAL PROOF:</u>

For 1st circuit:


SONY
HK
POWER
BATTERY

For 2nd circuit:



<u>Using De-Morgan's theorem to prove answer algebraically:</u>

Taking expression of 2nd circuit, and applying de-morgan's 1st theorem

$$\overline{((\bar{A}.B)+C)} = (\overline{(\bar{A}.B)}.\bar{C})$$

$$= (A+\bar{B}).\bar{C}$$

Thus, the expression for 2nd circuit when simplified using de-morgan's theorem is equal to the 1st circuit.

PROOF USING TRUTH TABLE METHOD:

A	B	C	A'	B'	C'	A+B'	(A+B').C'	A'.B	(A'.B)+C	((A'.B)+C)'
0	0	0	1	1	1	1	1	0	0	1
0	1	0	1	0	1	0	0	1	1	0
1	0	0	0	1	1	1	1	0	0	1
1	1	0	0	0	1	1	1	0	0	1
0	0	1	1	1	0	1	0	0	1	0
0	1	1	1	0	0	0	0	1	1	0
1	0	1	0	1	0	1	0	0	1	0
1	1	1	0	0	0	1	0	0	1	0

CONCLUSION:

- Using de-morgan's theorem, we can make cost-effective circuits by simplifying complex circuits.